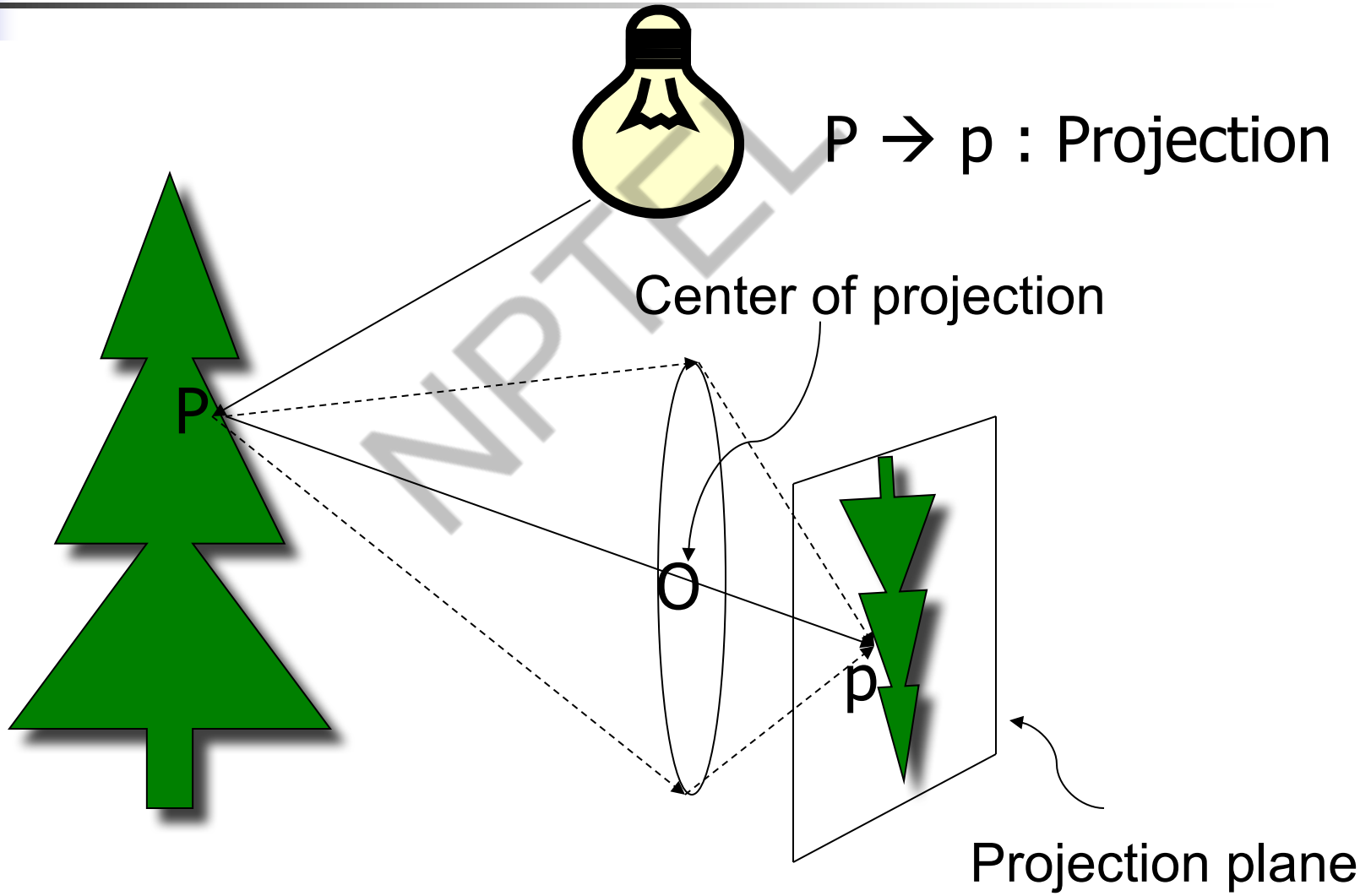
Projective Geometry-I
Week-02-L5

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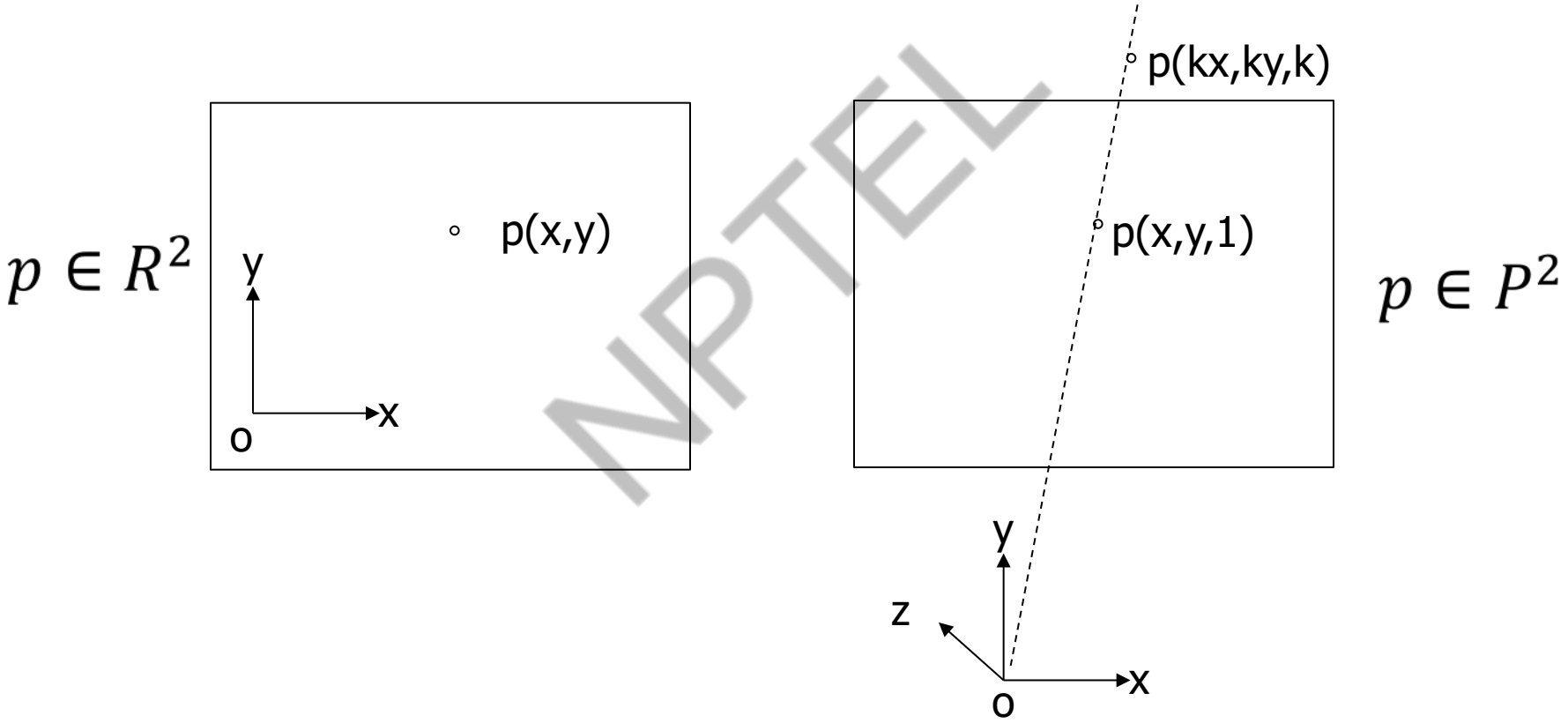
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Image formation in optical camera



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Real Space and Projective Space (2D)



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p(x,y) ⇔ p(kx,ky,k)

Homogeneous Coordinate system

Homogeneous Representation

A point in R^2: $\vec{x} \equiv \begin{bmatrix} x \\ y \end{bmatrix}$ $\Leftrightarrow$ A point in P^2: $\vec{X} \equiv \begin{bmatrix} kx \\ ky \\ k \end{bmatrix}$

$$P^2 = R^3 - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Singular point in the projective space.

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Homogeneous Representation

$$\text{In } R^2\text{: } ? \quad \Leftrightarrow \quad \text{In } P^2 : \vec{X} \equiv \begin{bmatrix} 25 \\ 30 \\ 5 \end{bmatrix}$$

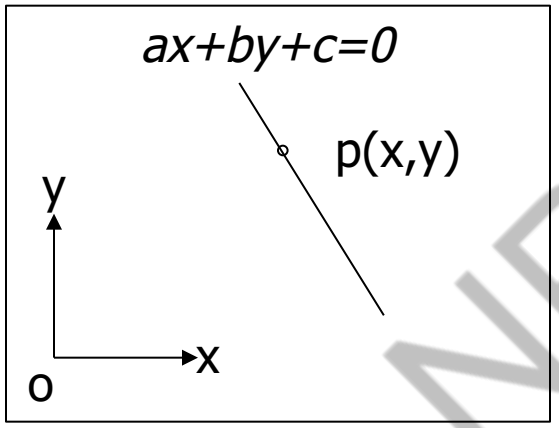
$$\text{The point in } R^2\text{: } \vec{x} \equiv \begin{bmatrix} \frac{25}{5} \\ \frac{30}{5} \end{bmatrix} \equiv \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\text{In } P^2 : \vec{X} \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \Leftrightarrow \quad \text{In } R^2\text{: } ?$$

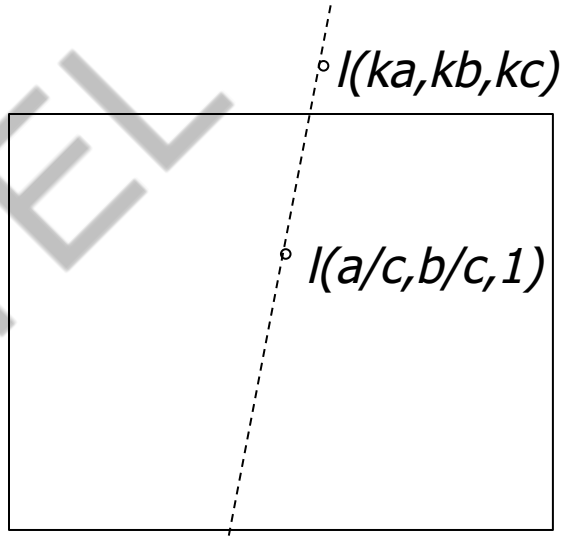
Does not belong to P^2.

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Homogeneous representation of a line in a plane



$p \in R^2$



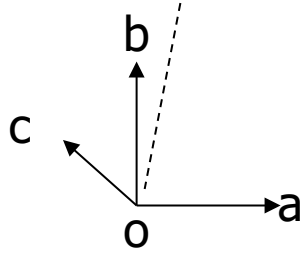
$l \in P^2$

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A point in P^2

A line in P^2

$$[x \quad y \quad 1]\begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

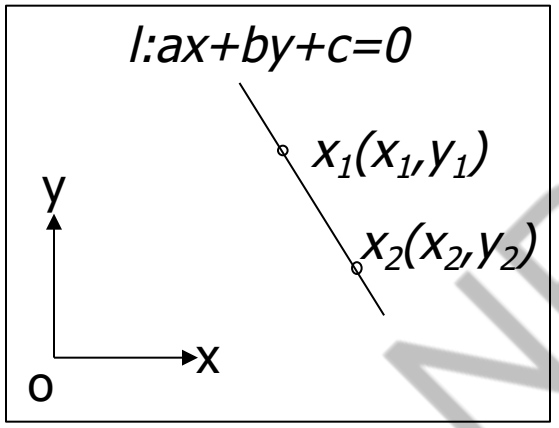


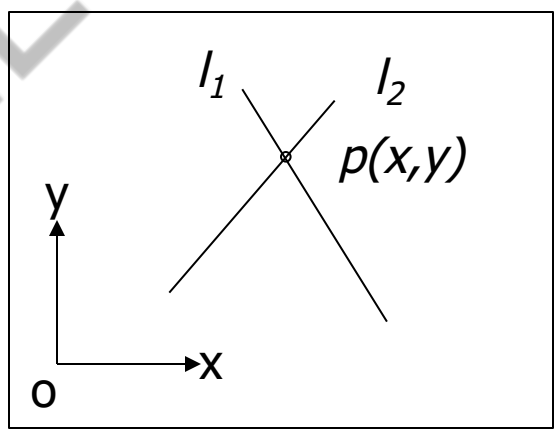
Point containment in P^2 $\Rightarrow$ $\vec{X}^T.\vec{l}=0 \Leftrightarrow \vec{l}^T.\vec{X}=0$

$$X_1 \times X_2 = \begin{vmatrix} i & j & k \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix}$$

Points and lines in P^2

$x_1 \in R^2$





$$\vec{l} = \vec{X}_1 \times \vec{X}_2$$

$$\vec{P} = \vec{l}_1 \times \vec{l}_2$$

Exactly one line through two points.

Exactly one point at intersection of two lines.

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Examples

1. Compute the line passing through (3,5) and (5,0) in a plane.

$$\vec{l} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} \times \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -25 \end{bmatrix}$$

2. Compute the point of intersection of the lines: 5x-2y+4=0 and 6x-7y-3=0.

$$\vec{P} = \begin{bmatrix} 5 \\ -2 \\ 4 \end{bmatrix} \times \begin{bmatrix} 6 \\ -7 \\ -3 \end{bmatrix} = \begin{bmatrix} 34 \\ 35 \\ -23 \end{bmatrix} \Rightarrow \left(-\frac{34}{23}, -\frac{35}{23}\right)$$

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Duality

$$x \longleftrightarrow l$$

$$x^T l = 0 \longleftrightarrow l^T x = 0$$

$$x = l \times l' \longleftrightarrow l = x \times x'$$

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Duality principle:

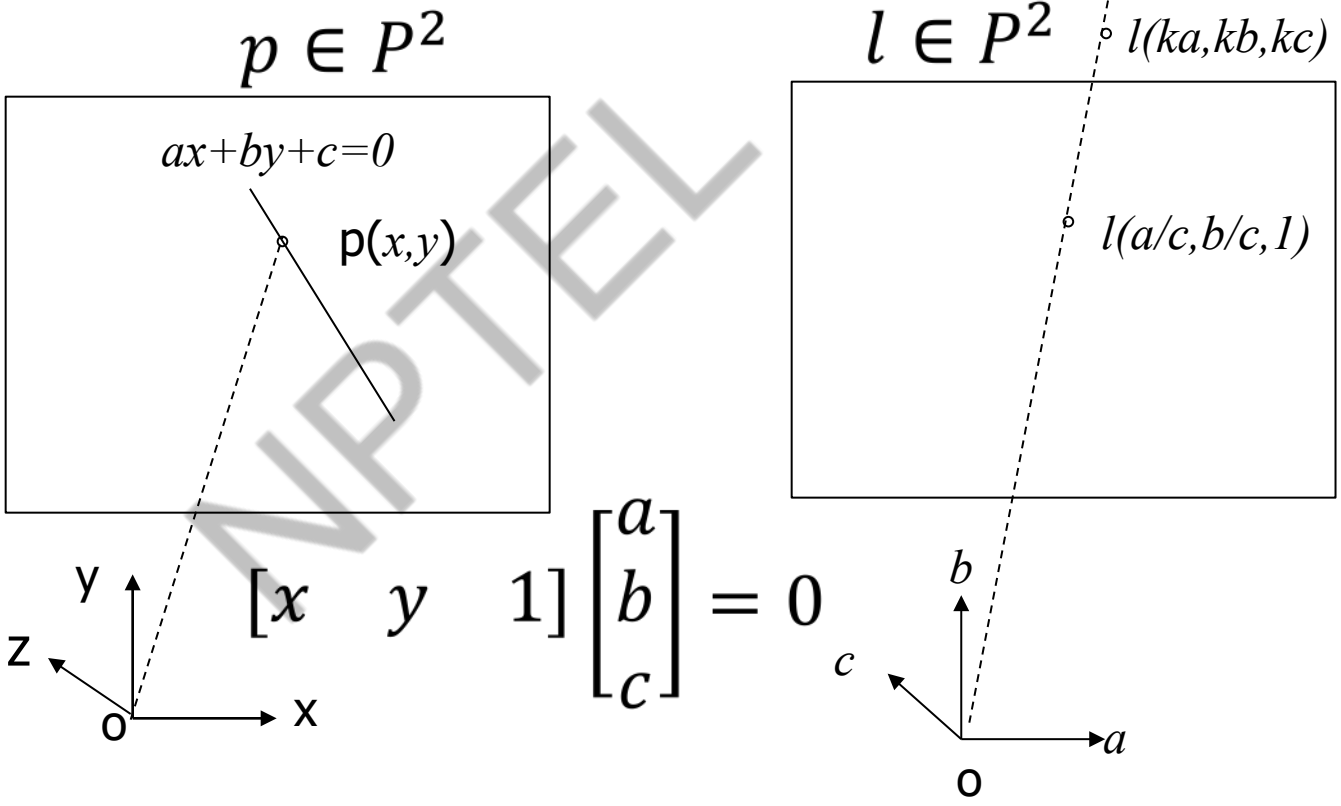
To any theorem of 2-dimensional projective geometry there corresponds a dual theorem, which may be derived by interchanging the role of points and lines in the original theorem.

Projective Geometry- Part II
Week 02 Lecture # 6

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Points and lines in a plane



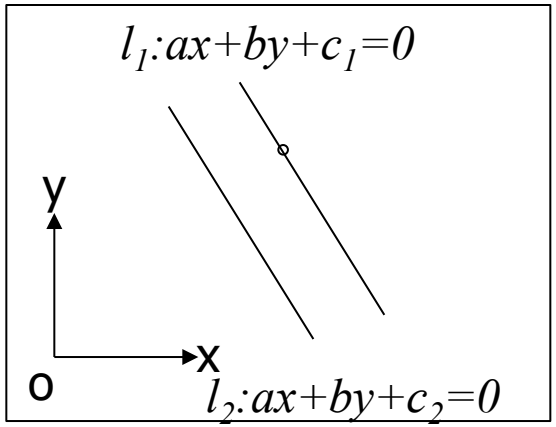
$$[x \quad y \quad 1]\begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$x^T l = 0 \longleftrightarrow l^T x = 0$$

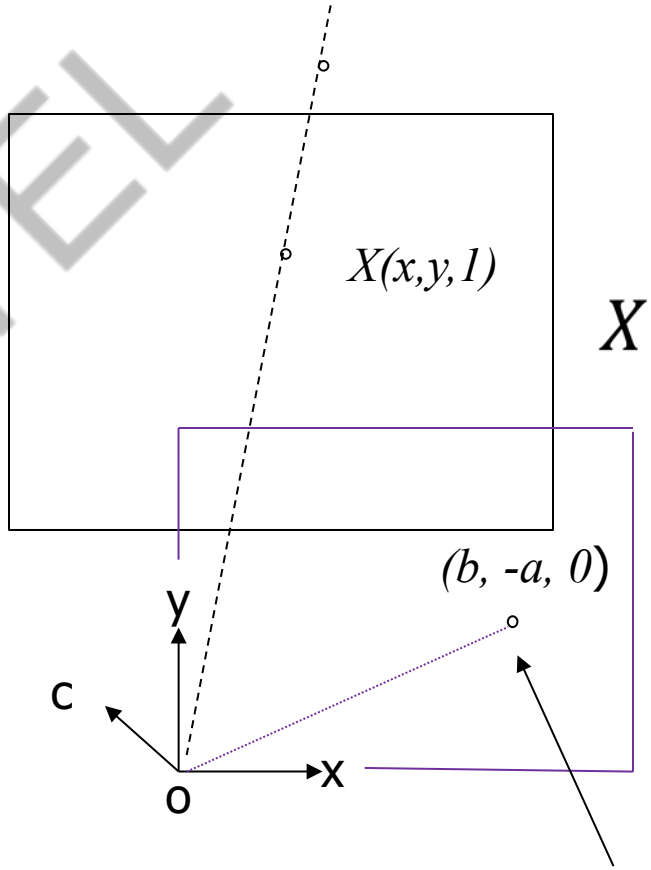
$$x = l \times l' \longleftrightarrow l = x \times x'$$

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Intersection of parallel lines



$$\vec{l}_1 \times \vec{l}_2 = (c_2 - c_1)\begin{bmatrix} b \\ -a \\ 0 \end{bmatrix}$$



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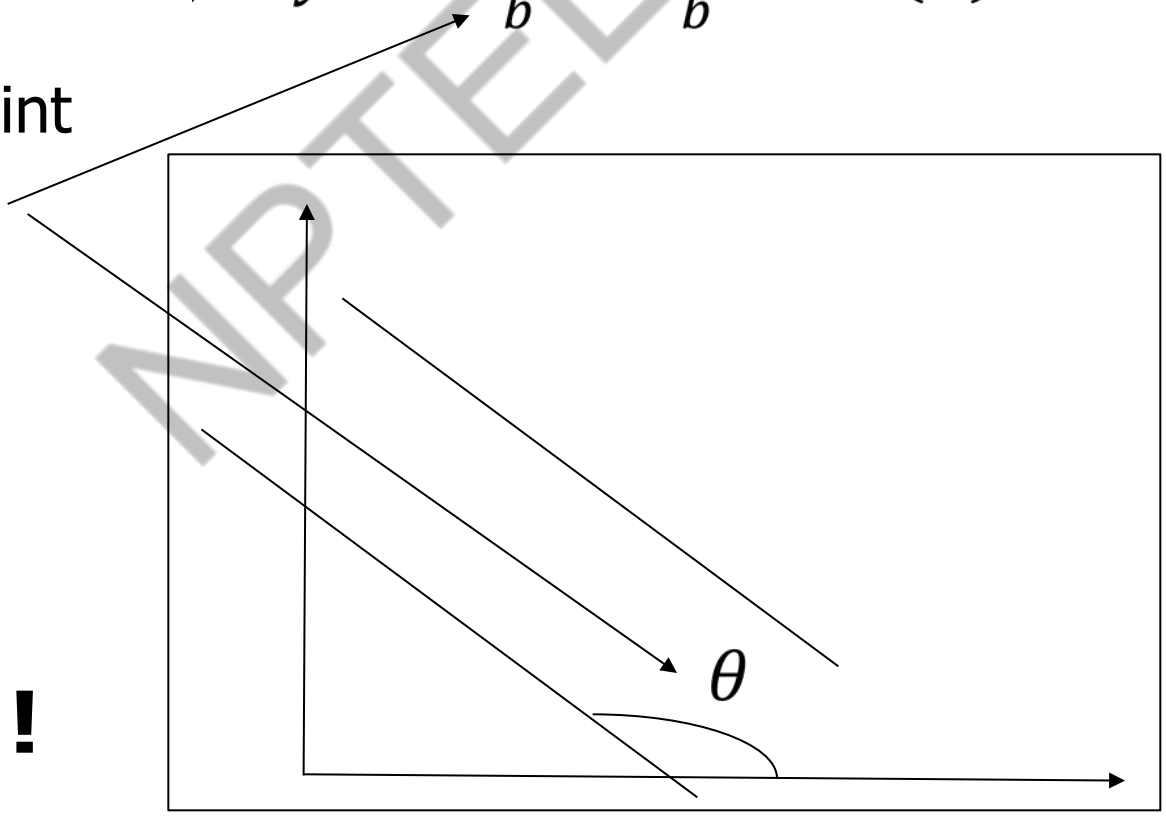
Meaning of an ideal point

$$ax + by + c = 0 \quad \Rightarrow \quad y = -\frac{a}{b}x + \frac{c}{b} = \tan(\theta)\, x + c'$$

Intersection point

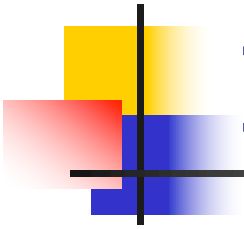
↓

$(b, -a, 0)$



A direction !

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Ideal points

Ideal points: Points on the X-Y plane or principal plane parallel to projection plane.

For canonical coordinate system, they are of the form:

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

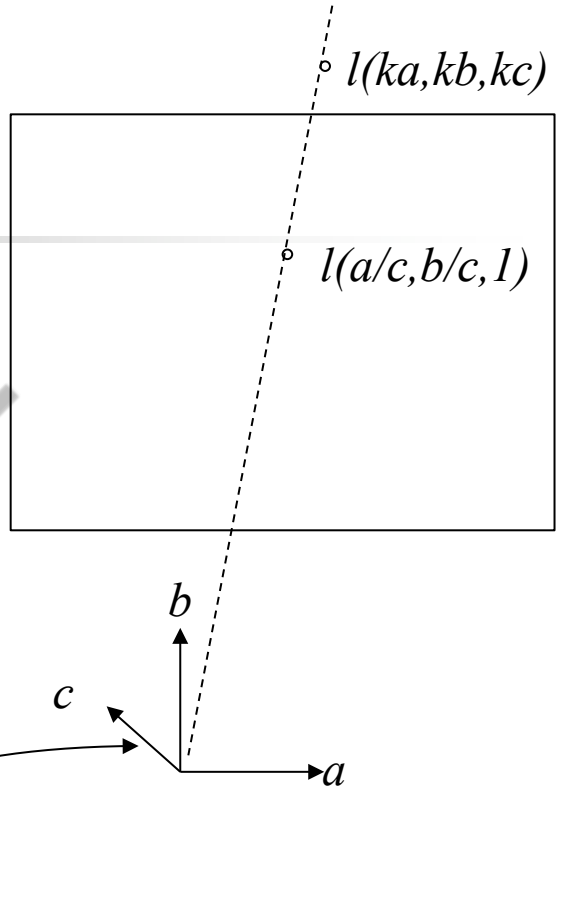
An ideal point denotes a direction toward infinity!

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Line at infinity

$$[x \quad y \quad 0]\begin{bmatrix}0\\0\\1\end{bmatrix} = 0$$

Any ideal point $\begin{bmatrix}0\\0\\1\end{bmatrix}$

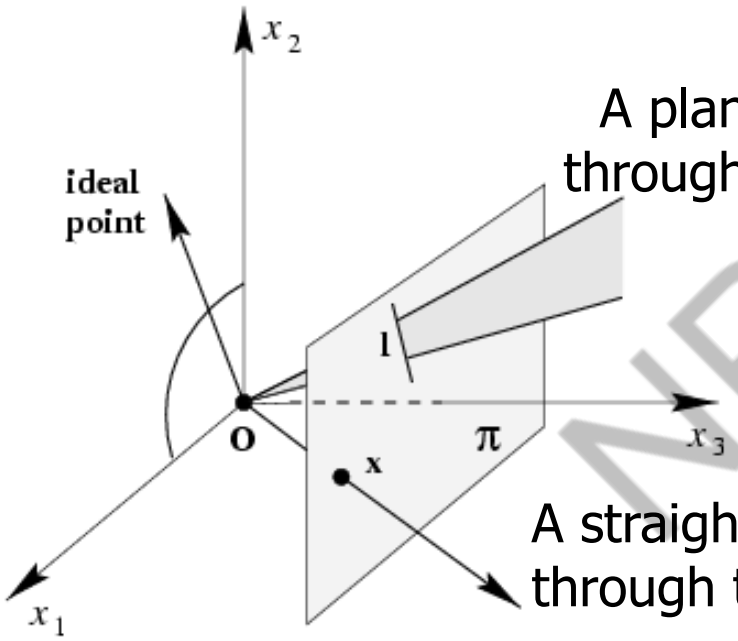


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Line at infinity (l_∞): Line containing every ideal point.

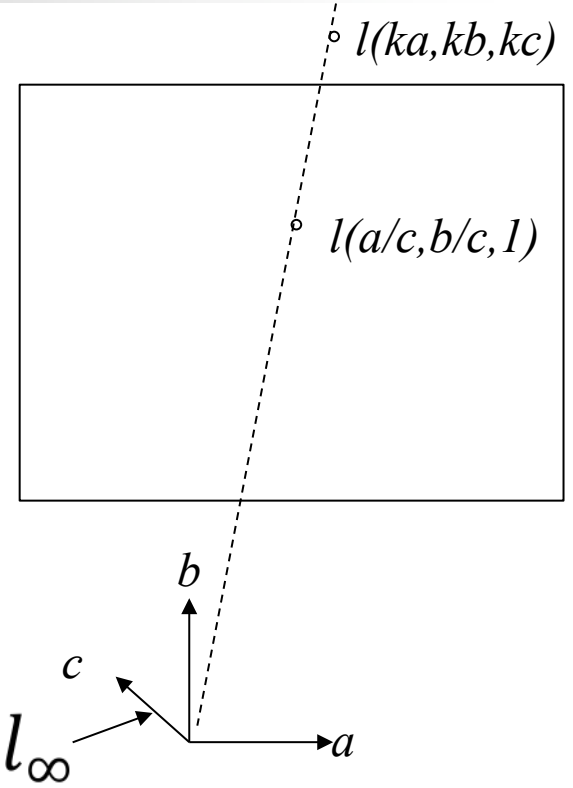
In canonical system, it is $\begin{bmatrix}0\\0\\1\end{bmatrix}$.

A model for the projective plane



A plane passing through the origin.

A straight line passing through the origin.



$$P^2 = R^3 - \begin{bmatrix}0\\0\\0\end{bmatrix} = R^2 \cup l_\infty$$

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press. (2000)

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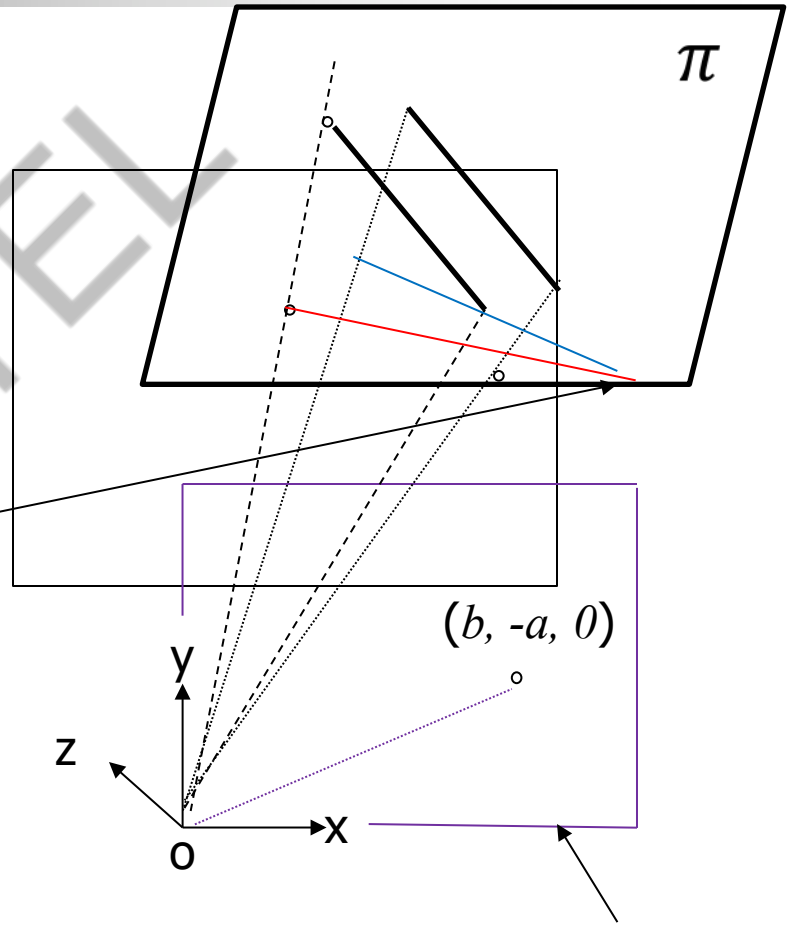
Projection of parallel lines from any arbitrary plane

Canonical projection plane (CPP)

Vanishing Point

⇩

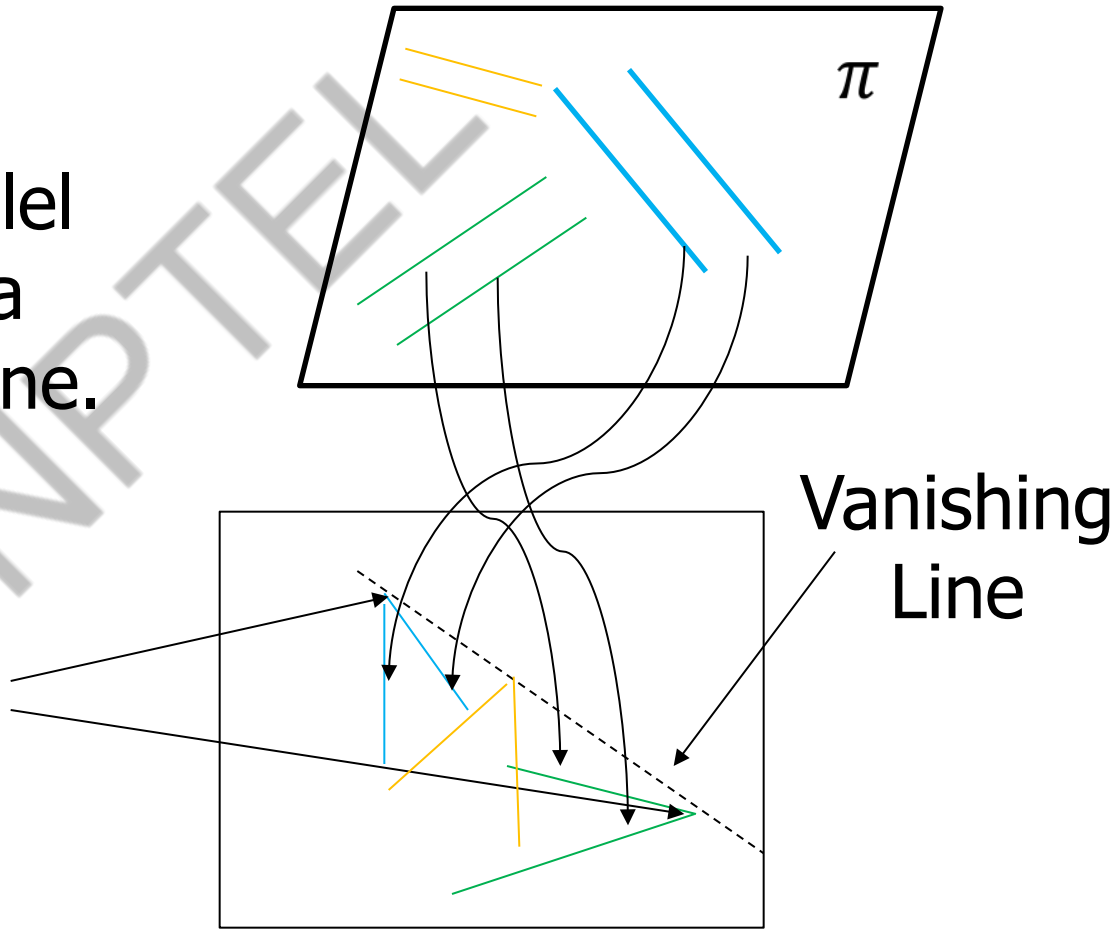
Point of intersection of parallel lines on π.



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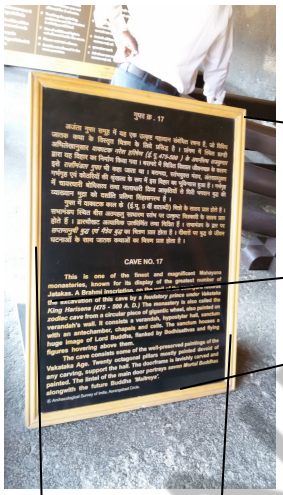
Vanishing points

Vanishing Points corresponding to parallel lines of a plane lie on a line, called vanishing line.



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A real life example



Vanishing points

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A journey toward infinity ….



Conics in P^2

- Curves described by 2nd degree equation in the plane.

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

- A point in homogeneous coordinate: (x_1, x_2, x_3)

$$\rightarrow (x_1/x_3, x_2/x_3)$$

$$a\left(\frac{x_1}{x_3}\right)^2 + \mathrm{b}\left(\frac{x_1}{x_3}\right)\left(\frac{x_2}{x_3}\right) + \mathrm{c}\left(\frac{x_2}{x_3}\right)^2 + \mathrm{d}\left(\frac{x_1}{x_3}\right) + \mathrm{e}\left(\frac{x_2}{x_3}\right) + \mathrm{f} = 0$$

$$\Longrightarrow ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

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Conics in P^2

$$ax_1^2 + bx_1x_2 + cx_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$$

$$\Rightarrow X^T C X = 0$$

Where

$$C = \begin{bmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{bmatrix}$$

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Conics identified by C with 5 d.o.f. (a:b:c:d:e:f)

Five points define a conic

For each point the conic passes through

$$ax_i^2 + bx_i y_i + cy_i^2 + dx_i + ey_i + f = 0$$

$$\Rightarrow \quad \left(x_i^2, x_i y_i, y_i^2, x_i, y_i, f\right)\mathbf{c} = 0$$

$$\mathbf{c} = \left(a, b, c, d, e, f\right)^{\mathsf{T}}$$

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Five points define a conic

For each point the conic passes through

$$ax_i^2 + bx_i y_i + cy_i^2 + dx_i + ey_i + f = 0$$

Stacking constraints yields

$$\begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \mathbf{c} = 0$$

- Rank deficient **C**

→ degenerate conic

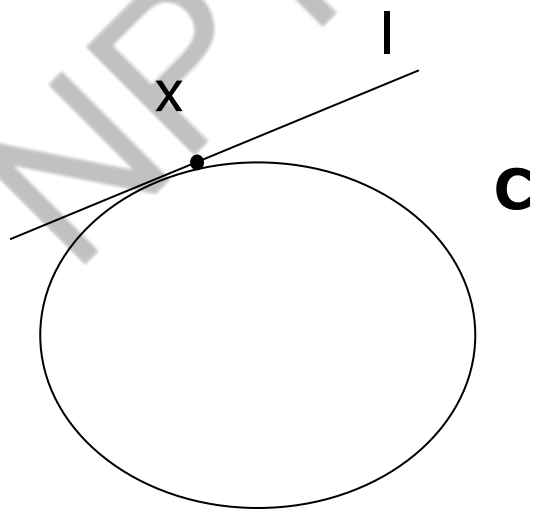
two lines (of rank 2)

a repeated line (of rank 1).

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Tangent lines to conics

The line **l** tangent to **C** at point x on **C** is given by **l**=**C**x



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From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Dual conics

A line tangent to the conic **C** satisfies $l^T C^* l = 0$

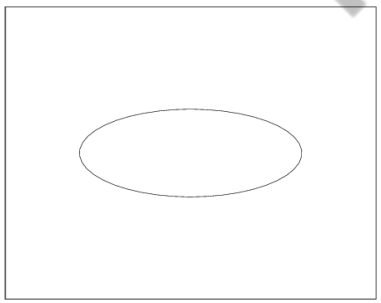
As **C** is symmetric, C^{-1}

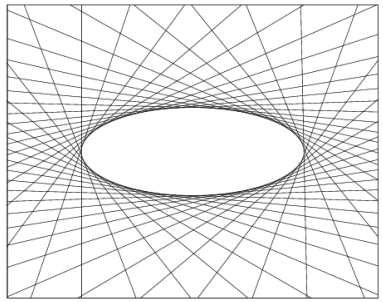
$$\mathbf{l} = \mathbf{Cx} \Rightarrow \mathbf{x} = \mathbf{C^{-1}l}$$

$$\mathbf{x^T Cx} = 0 \Rightarrow \left(\mathbf{C^{-1}l}\right)^{\mathbf{T}} \mathbf{C}\left(\mathbf{C^{-1}l}\right) = 0$$

$$\Rightarrow \mathbf{l^T}\left(\mathbf{C^{-1}}\right)^{\mathbf{T}} \mathbf{CC^{-1}l} = 0 \Rightarrow \mathbf{l^T C^{-T} l} = 0$$

Dual conics = line conics = conic envelopes





From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

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Degenerate Conics

- Rank of C <3
- Rank 2 → Two lines / points
- Rank 1 → One repeated lines / points
- Degenerate point conic:

  $\mathbf{C}=\mathbf{l}.\mathbf{m}^T+\mathbf{m}.\mathbf{l}^T$ rank 2, if $\mathbf{l} <> \mathbf{m}$
- Degenerate dual line conic:

  $\mathbf{C}^*=\mathbf{x}.\mathbf{y}^T+\mathbf{y}.\mathbf{x}^T$ rank 2, if $\mathbf{x} <> \mathbf{y}$

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- $\mathbf{x}^T\mathbf{l}=0$, and $\mathbf{l}^T\mathbf{x}=0$
- $\mathbf{x}=\mathbf{l} \times \mathbf{l}'$, and $\mathbf{l}=\mathbf{x} \times \mathbf{x}'$

Summary

- A point in a 2-D projective space represents a ray passing through origin of an implicit 3D space.
  - Requires additional dimension for representation.
    - Homogeneous Coordinate Representation
- Straight lines in R^2 are elements of a 2D projective space.
- Points and lines hold duality theorem.
- Conics are represented by a 3x3 symmetric matrix.
  - Every conic has a dual conic or line conic as an envelop of its tangents.

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Projective Transformation
Week-02 Lecture # 7

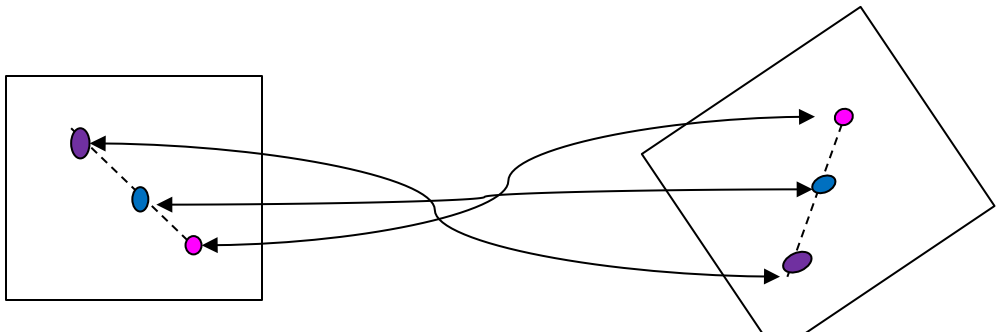
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Projective transformation

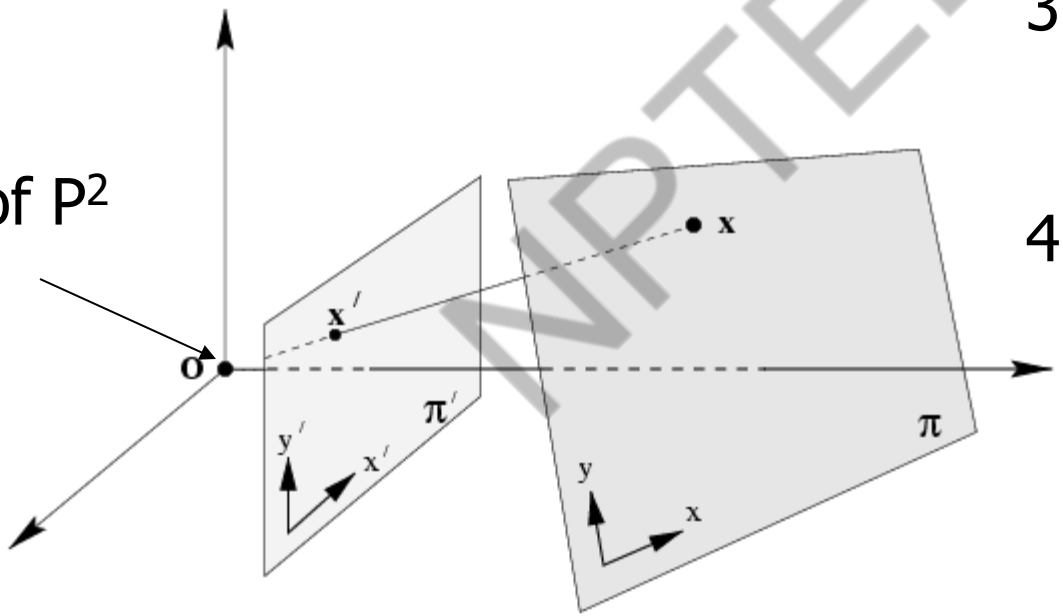
- h: $P^2 \rightarrow P^2$
- Invertible
- Collinearity of every three points to be preserved, i.e. three points x_1, x_2, x_3 lie on the same line if and only if $h(x_1), h(x_2), h(x_3)$ do.

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An example: change of coordinate convention

Origin of P^2 fixed.



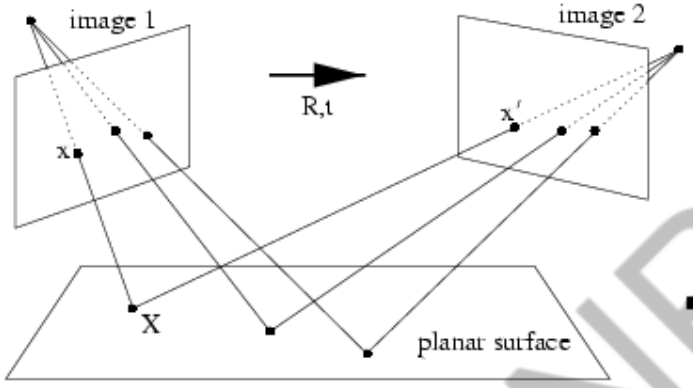
1. Rotation of axes.
2. Change of scale.
3. Translation of origin in planar coordinate system.
4. Use of Affine coordinate system in plane.

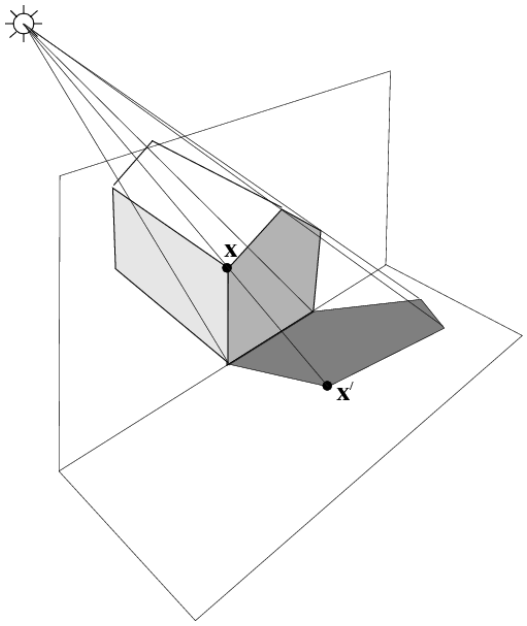
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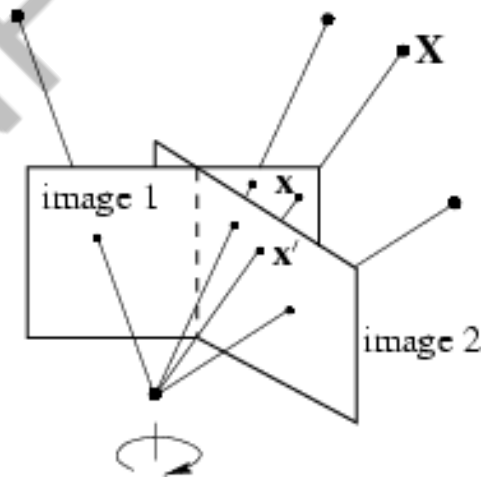
From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

More examples

Shift of origin of P^2







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From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Form of h

- Only one form possible.
- It is linear and invertible.

$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

8DOF

$$\mathbf{X}' = \mathbf{H}\mathbf{X} \equiv k\mathbf{H}\mathbf{X}$$

Also called homography and **H** is the homography matrix.

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Hx preserves collinearity.

- Let $\boldsymbol{l}$ be a line in P^2.
- A point $\mathbf{x}$ on $\boldsymbol{l}$ satisfies

$$\boldsymbol{l}^T\mathbf{x}=\mathbf{0}$$

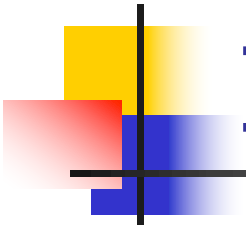
$$\rightarrow \boldsymbol{l}^T\mathbf{H}^{-1}\mathbf{H}\mathbf{x}=\mathbf{0}$$

$$\rightarrow (\mathbf{H}^{-T}\boldsymbol{l})^T\mathbf{x}=\mathbf{0}$$

- $\mathbf{H}^{-T}\boldsymbol{l}$ is the transformed line of $\boldsymbol{l}$.

Harder to show that **H** is the only form of homography.

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Implications

- If there is a homography, there exists a unique **H**, which is a 3x3 invertible matrix.
- Functional form known, so easier to estimate.
- **H** and k**H** are equivalent, where k is a scalar constant.
- Number of unknowns in **H** = 8.

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Estimation of **H**

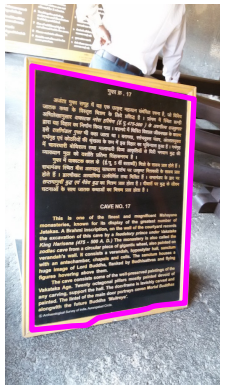
- Given point correspondences ($\mathbf{x_i}$, $\mathbf{x_i'}$) estimate **H** such that $\mathbf{x_i'} = \mathbf{H}\mathbf{x_i}$.
- There are 8 unknowns.
- $\mathbf{x'} = \mathbf{H}\mathbf{x}$ → Two independent equations.

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}} \qquad y' = \frac{x'_2}{x'_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

- Minimum 4 point correspondences needed.

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Removing projective distortion





1. Select four points in a plane with known coordinates.
2. Form equations.

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{x'_2}{x'_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

(linear in h_{ij})

$$x'\left(h_{31}x + h_{32}y + h_{33}\right) = h_{11}x + h_{12}y + h_{13}$$

$$y'\left(h_{31}x + h_{32}y + h_{33}\right) = h_{21}x + h_{22}y + h_{23}$$

Remark: no calibration at all necessary.
Does not work if $h_{33}=0$ in **H**.

3. Setting h_{33} at *1* solve them.

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Form equations

$$x'(h_{31}x + h_{32}y + h_{33}) = h_{11}x + h_{12}y + h_{13}$$
$$y'(h_{31}x + h_{32}y + h_{33}) = h_{21}x + h_{22}y + h_{23}$$

(51,791) →(1, 900)
(63, 143) → (1,1)
(444 ,211) → (501,1)
(426, 719) → (501,900)

$$-51\,h_{11} - 791\,h_{12} - h_{13} + 51h_{31} + 791h_{32} = -1$$
$$-51\,h_{21} - 791\,h_{22} - h_{23} + 45900\,h_{31} + 711900\,h_{32} = -900$$
$$-63\,h_{11} - 143\,h_{12} - h_{13} + 63h_{31} + 43\,h_{32} = -1$$
$$-63\,h_{21} - 143\,h_{22} - h_{23} + 63\,h_{31} + 143\,h_{32} = -1$$
$$-444h_{11} - 211h_{12} - h_{13} + 222444\,h_{31} + 105711h_{32} = -501$$
$$-444h_{21} - 211\,h_{22} - h_{23} - 444\,h_{31} + 211\,h_{32} = -1$$
$$-426h_{11} - 719\,h_{12} - h_{13} + 213426\,h_{31} + 360219\,h_{32} = -501$$
$$-426h_{21} - 719\,h_{22} - h_{23} + 383400\,h_{31} + 647100\,h_{32} = -900$$

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In matrix form

$$\begin{bmatrix} -51 & -791 & -1 & 0 & 0 & 0 & 51 & 791 \\ 0 & 0 & 0 & -51 & -791 & 1 & 45900 & 711900 \\ -63 & -143 & -1 & 0 & 0 & 0 & 63 & 143 \\ 0 & 0 & 0 & -63 & -143 & -1 & 63 & 143 \\ -444 & -211 & -1 & 0 & 0 & 0 & 222444 & 105711 \\ 0 & 0 & 0 & -444 & -211 & -1 & 444 & 211 \\ -426 & -719 & -1 & 0 & 0 & 0 & 213426 & 360219 \\ 0 & 0 & 0 & -426 & -719 & -1 & 383400 & 647100 \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} -1 \\ -900 \\ -1 \\ -1 \\ -501 \\ -1 \\ -501 \\ -900 \end{bmatrix}$$

NPTEL

Solve by matrix inversion. $h_{33}=1$

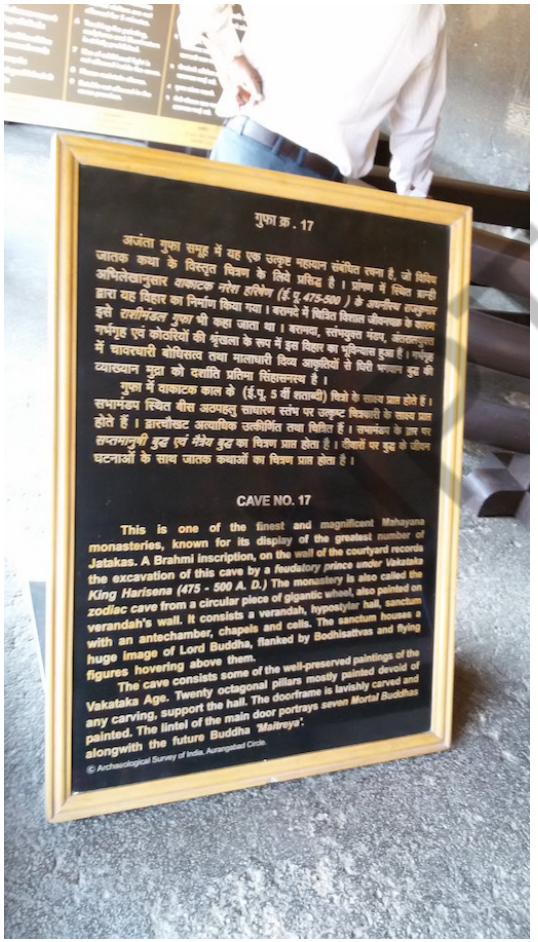
$$H = \begin{bmatrix} 0.9791 & 0.0181 & -63.3104 \\ -0.2303 & 1.2874 & -168.6295 \\ -0.0005 & -0.0001 & 1.0000 \end{bmatrix}$$

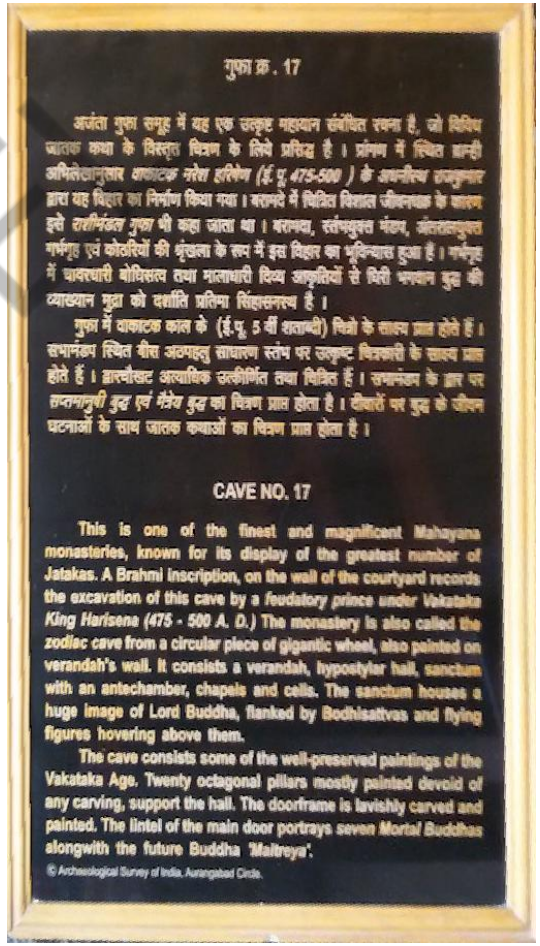
Solve equations

$$H = \begin{bmatrix} 0.9791 & 0.0181 & -63.3104 \\ -0.2303 & 1.2874 & -168.6295 \\ -0.0005 & -0.0001 & 1.0000 \end{bmatrix}$$

NPTEL

Apply homography





Direct Linear Transformation (DLT)

$$\mathbf{x}_i' = (x_i', y_i', w_i')^{\mathsf{T}}$$

$$H = \begin{bmatrix} h^{1T} \\ h^{2T} \\ h^{3T} \end{bmatrix}$$

$$\mathbf{x}_i' = \mathrm{H}\mathbf{x}_i$$

$$\mathbf{x}_i' \times \mathrm{H}\mathbf{x}_i = 0$$

$$\mathrm{H}\mathbf{x}_i = \begin{pmatrix} \mathrm{h}^{1\mathsf{T}}\mathbf{x}_i \\ \mathrm{h}^{2\mathsf{T}}\mathbf{x}_i \\ \mathrm{h}^{3\mathsf{T}}\mathbf{x}_i \end{pmatrix}$$

$$\mathbf{x}_i' \times \mathrm{H}\mathbf{x}_i = \begin{pmatrix} y_i'\mathrm{h}^{3\mathsf{T}}\mathbf{x}_i - w_i'\mathrm{h}^{2\mathsf{T}}\mathbf{x}_i \\ w_i'\mathrm{h}^{1\mathsf{T}}\mathbf{x}_i - x_i'\mathrm{h}^{3\mathsf{T}}\mathbf{x}_i \\ x_i'\mathrm{h}^{2\mathsf{T}}\mathbf{x}_i - y_i'\mathrm{h}^{1\mathsf{T}}\mathbf{x}_i \end{pmatrix} = 0$$

Redundant: x_i':(1)+y_i': (2)=(3)

NPTEL

Direct Linear Transformation (DLT)

$$\begin{bmatrix} 0^{\mathsf{T}} & -w'_i \mathbf{x}_i^{\mathsf{T}} & y'_i \mathbf{x}_i^{\mathsf{T}} \\ w'_i \mathbf{x}_i^{\mathsf{T}} & 0^{\mathsf{T}} & -x'_i \mathbf{x}_i^{\mathsf{T}} \\ -y'_i \mathbf{x}_i^{\mathsf{T}} & x'_i \mathbf{x}_i^{\mathsf{T}} & 0^{\mathsf{T}} \end{bmatrix} \begin{pmatrix} \mathbf{h}^1 \\ \mathbf{h}^2 \\ \mathbf{h}^3 \end{pmatrix} = 0$$

$\mathbf{A}_i \mathbf{h} = 0$ where $A_i = \begin{bmatrix} 0^T & -w'_i x_i^T & y'_i x_i^T \\ w'_i x_i^T & 0^T & -x'_i x_i^T \end{bmatrix}$

Dimension of A_i : 2 x 9.

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Direct Linear Transformation (DLT): Non-homogeneous Equations

- Solving for H by setting $h_{33}=1$.

$$h = \begin{bmatrix} \tilde{h} \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & -x_i w'_i & -y_i w'_i & -w_i w'_i & x_i y'_i & y_i y'_i \\ x_i w'_i & y_i w'_i & w_i w'_i & 0 & 0 & 0 & -x_i x'_i & -y_i x'_i \end{bmatrix} \tilde{h} = \begin{bmatrix} -w_i y'_i \\ w_i x'_i \end{bmatrix}$$

$$\tilde{A}_i \tilde{h} = b_i$$

$$A\tilde{h} = b$$

$$Minimize \ \|A\tilde{h} - b\|$$

Solution: $\tilde{h} = (A^T A)^{-1} A^T b$

Dimension of *A*: 2nx8

Rank: 8

Dimension of *h*: 8x1

Dimension of *b*: 2nx1

Caution: If $h_{33}=0$, no multiplication scale exists, and no solution obtained.

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Direct Linear Transformation (DLT): Homogeneous Equations

- Solving for H: $\mathrm{Ah} = 0$

$$\mathrm{A} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ \vdots \\ A_n \end{bmatrix}$$

Dimension of *A*: 2nx9

Rank: 8

Dimension of *h*: 9x1

Dimension of *Ah*: 2nx1

Minimize $\|Ah\|$ such that $\|\mathrm{h}\| = 1$

Solution: Unit eigen vector of smallest eigen value of A^TA.

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Other error criteria

- Algebraic error: Error term in DLT.
- Geometric error: $\sum d_e^2\left(\mathbf{x}', \mathbf{Hx}\right)$ Euclidean distance
- Geometric error with reprojection:

$$\sum \left(d_e^2\left(\mathbf{x}', \mathbf{Hx}\right) + d_e^2\left(\mathbf{H^{-1}x}', \mathbf{x}\right)\right)$$

- Use of nonlinear iterative optimization techniques such as Newton iteration, Levenberg-Marquardt (LM) method, etc.

NPTEL

Transformation invariance and normalization

- Problem: To estimate **H** given a set of ($\mathbf{x_i}$, $\mathbf{x_i'}$) .
- Consider, $\mathbf{y_i}=\mathbf{T}\mathbf{x_i}$ and $\mathbf{y'_i}=\mathbf{T'}\mathbf{x_i'}$ for known **T** and **T'**, which are invertible.
- Now estimate homography **G** from ($\mathbf{y_i}$, $\mathbf{y'_i}$).
- Can you estimate **H** from **G**?

$$\mathbf{x}' = \mathbf{H}\mathbf{x}$$
$$\Rightarrow \mathbf{T}'^{-1}\mathbf{y}' = \mathbf{H}\mathbf{T}^{-1}\mathbf{y}$$
$$\Rightarrow \mathbf{y}' = \mathbf{T}'\mathbf{H}\mathbf{T}^{-1}\,\mathbf{y}$$

G

Caution: For DLT it is not equivalent.
As the constraint ||g||=1 is not equivalent to ||h||=1.

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Robust computation through Normalization of data

- Transform the point set so that its center becomes origin (in the plane) and avg. distance from it is $\sqrt{2}$.

$$x_i^{(n)} = \frac{x_i - \bar{x}}{\sigma_x} \qquad y_i^{(n)} = \frac{y_i - \bar{y}}{\sigma_y}$$

- Apply DLT on transformed point.
- Recover homography from the homography of transformed point sets.

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Summary

- A projective transformation, invertible and preserving collinearity, is always in a linear form.
  - $\mathbf{x}' = \mathbf{H}\mathbf{x}$
  - $\mathbf{l}' = \mathbf{H}^{-T}\mathbf{l}$
- A computational problem: estimation of homography given a set of point correspondences.
  - Minimum 4 point correspondences needed.
  - Direct linear transformation (using LSE).
  - Use of linear transformation of points to make the computation robust.

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Homography: Properties
Week 02 Lectures 8-10

NPTEL

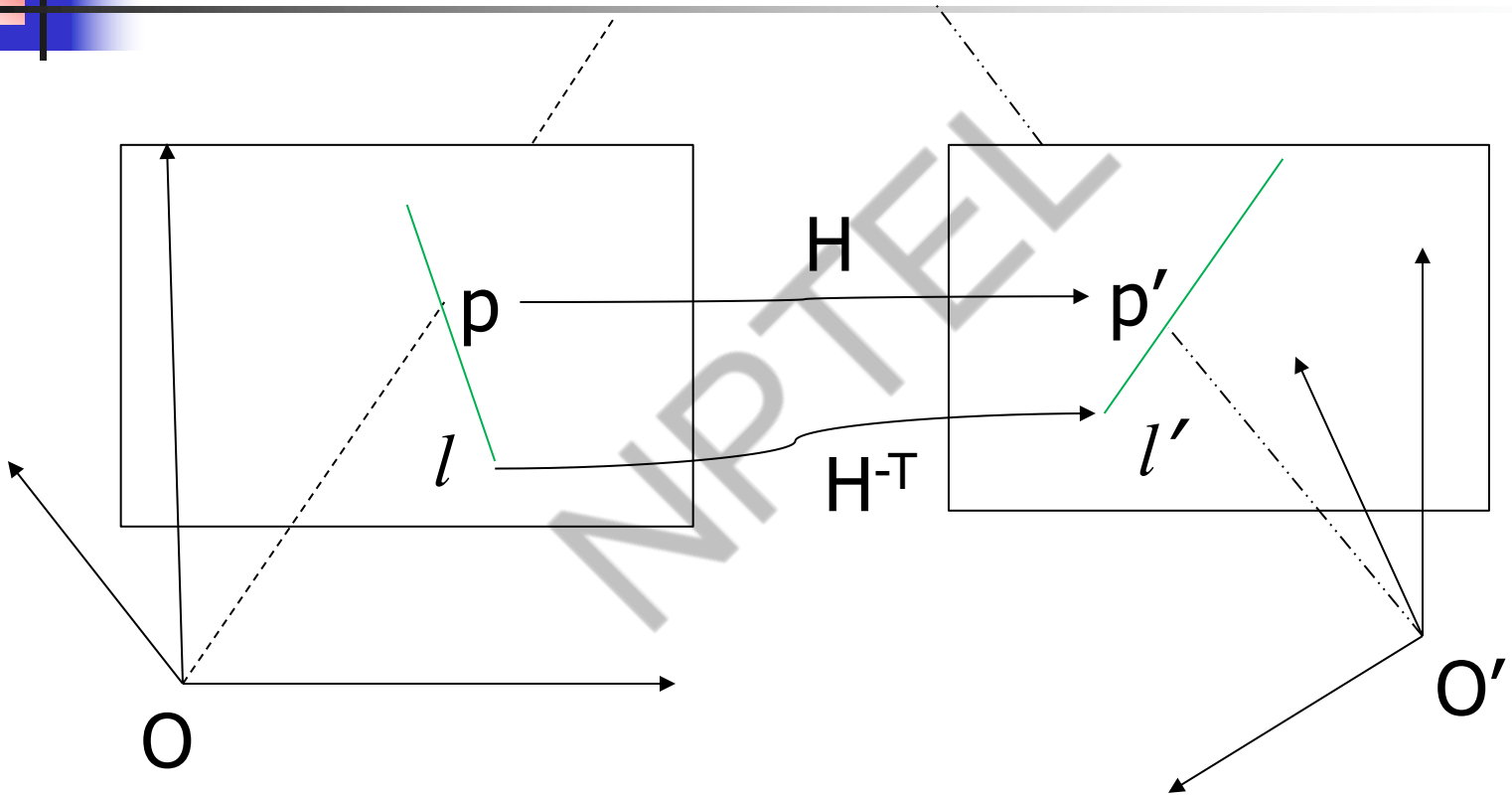
Jayanta Mukhopadhyay
Dept. of Computer Science and Engg.

Projective transformation

- h: $P^2 \rightarrow P^2$.
- Invertible.
- Collinearity of every three points to be preserved, i.e. three points x_1, x_2, x_3 lie on the same line if and only if $h(x_1), h(x_2), h(x_3)$ do.
- Only in the form of non-singular 3x3 matrix.

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$P^2 \rightarrow P^2$



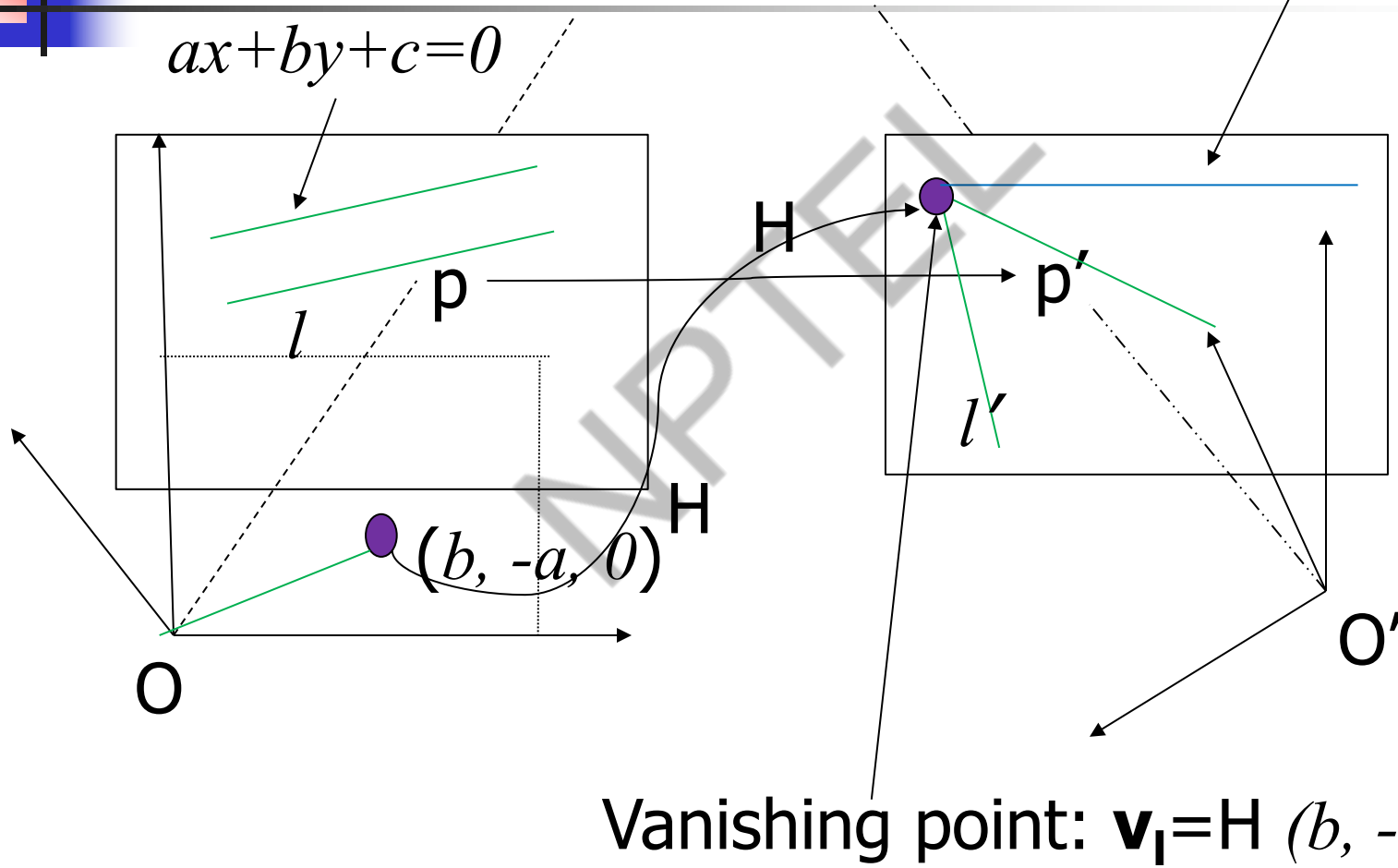
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Point: $\mathbf{x'=Hx}$

Line: $\mathbf{l'=H^{-T}l}$

Vanishing line=$H^{-T}\mathbf{l}_\alpha = H^{-T}(0, 0, 1)^T$

Vanishing point and line



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Vanishing point: $\mathbf{v}_l = H\,(b, -a, 0)^T$

Point and line transformation

- Point transformation:
  - $\mathbf{x'}=\mathbf{Hx}$
- Line transformation:
  - $\mathbf{l'}=\mathbf{H}^{-T}\mathbf{l}$
- Vanishing point for lines parallel to $\mathbf{l}=(a,b,c)^T$:
  - $\mathbf{v_l}=\mathbf{H}\ (b,\ -a,\ 0)^T$
- Vanishing line:
  - $\mathbf{l_H}=\mathbf{H}^{-T}\mathbf{l}_\alpha$
    $=\mathbf{H}^{-T}\ (0,\ 0,1)^T$

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Examples

- Consider the following homography H between two projective spaces.

$$H=\begin{bmatrix} 3 & 4 & -6 \\ 1 & 3 & -8 \\ 0 & 5 & 1 \end{bmatrix}$$

Compute the transformation of the line formed by two points (2,4,2) and (6,9,3) in P^2.

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Method-I

$$H=\begin{bmatrix}3 & 4 & -6\\ 1 & 3 & -8\\ 0 & 5 & 1\end{bmatrix}$$

- Compute transformed points of (2,4,2) and (6,9,3).
- Take their cross product to compute the transformed line.

$$\begin{bmatrix}2\\4\\2\end{bmatrix} \equiv \begin{bmatrix}1\\2\\1\end{bmatrix} \qquad \begin{bmatrix}6\\9\\3\end{bmatrix} \equiv \begin{bmatrix}2\\3\\1\end{bmatrix}$$

$$H.\begin{bmatrix}1\\2\\1\end{bmatrix} = \begin{bmatrix}5\\-1\\11\end{bmatrix} \qquad H.\begin{bmatrix}2\\3\\1\end{bmatrix} = \begin{bmatrix}12\\3\\16\end{bmatrix}$$

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$$H=\begin{bmatrix}3 & 4 & -6\\ 1 & 3 & -8\\ 0 & 5 & 1\end{bmatrix}$$

Method-I

- Take their cross product to compute the transformed line.

$$H.\begin{bmatrix}1\\2\\1\end{bmatrix}=\begin{bmatrix}5\\-1\\11\end{bmatrix} \qquad H.\begin{bmatrix}2\\3\\1\end{bmatrix}=\begin{bmatrix}12\\3\\16\end{bmatrix}$$

$$\begin{bmatrix}5\\-1\\11\end{bmatrix}\times\begin{bmatrix}12\\3\\16\end{bmatrix}=\begin{bmatrix}-49\\52\\27\end{bmatrix}$$

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Method-II

Compute the line and transform it.

The line between the points l:
(1,2,1) x (2,3,1)→ (-1,1-1)

Transformed line: $l' = H^{-T} l$

$$H^{-1} = \frac{1}{95}\begin{bmatrix} 43 & -34 & -14 \\ -1 & 3 & 18 \\ 5 & -15 & 5 \end{bmatrix} \qquad l' = \frac{1}{95}\begin{bmatrix} -49 \\ 52 \\ 27 \end{bmatrix}$$

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$$H=\begin{bmatrix}3 & 4 & -6\\ 1 & 3 & -8\\ 0 & 5 & 1\end{bmatrix}$$

Example: Vanishing line

- Compute the vanishing line in the transformed space.

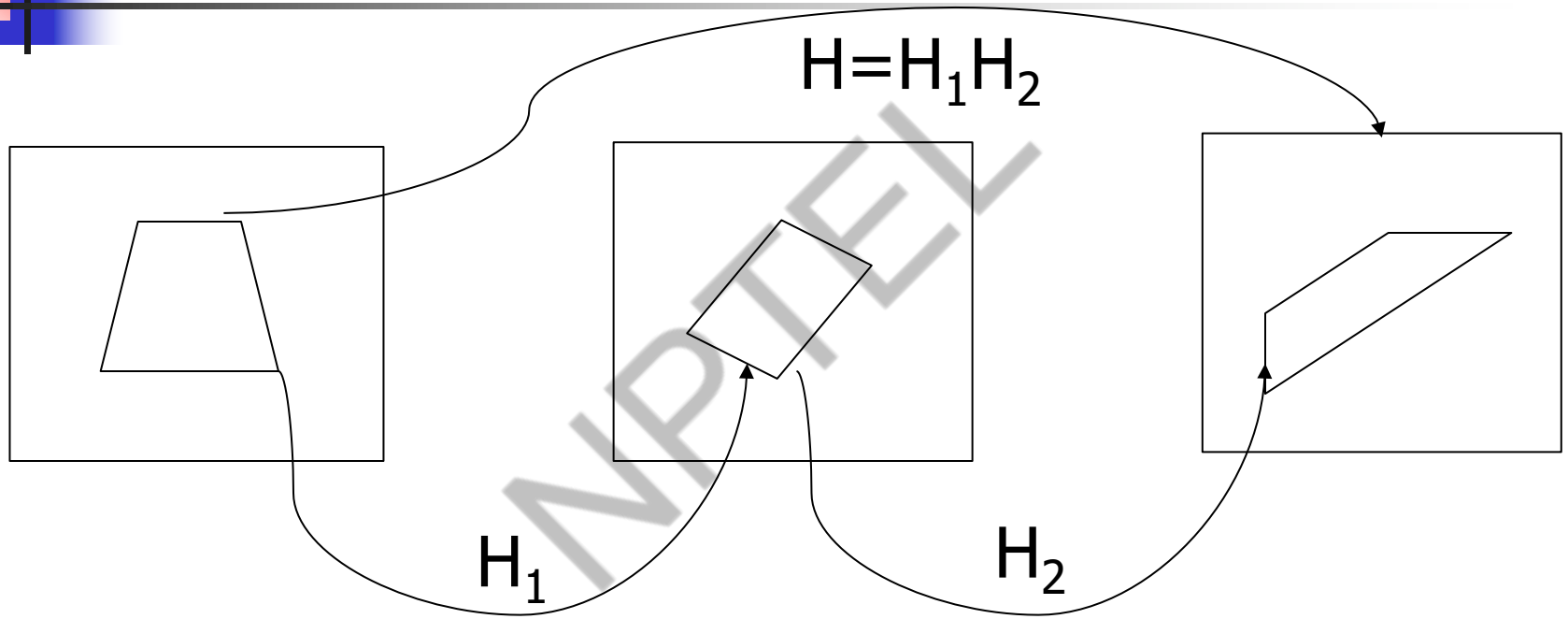
Transformed line: $l'_v = H^{-T} l_\alpha$

$$H^{-T} l_\infty = \frac{1}{95}\begin{bmatrix}43 & -1 & 5\\ -34 & 3 & -15\\ -14 & 18 & 5\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$l'_v = \begin{bmatrix}5\\-15\\5\end{bmatrix}$$

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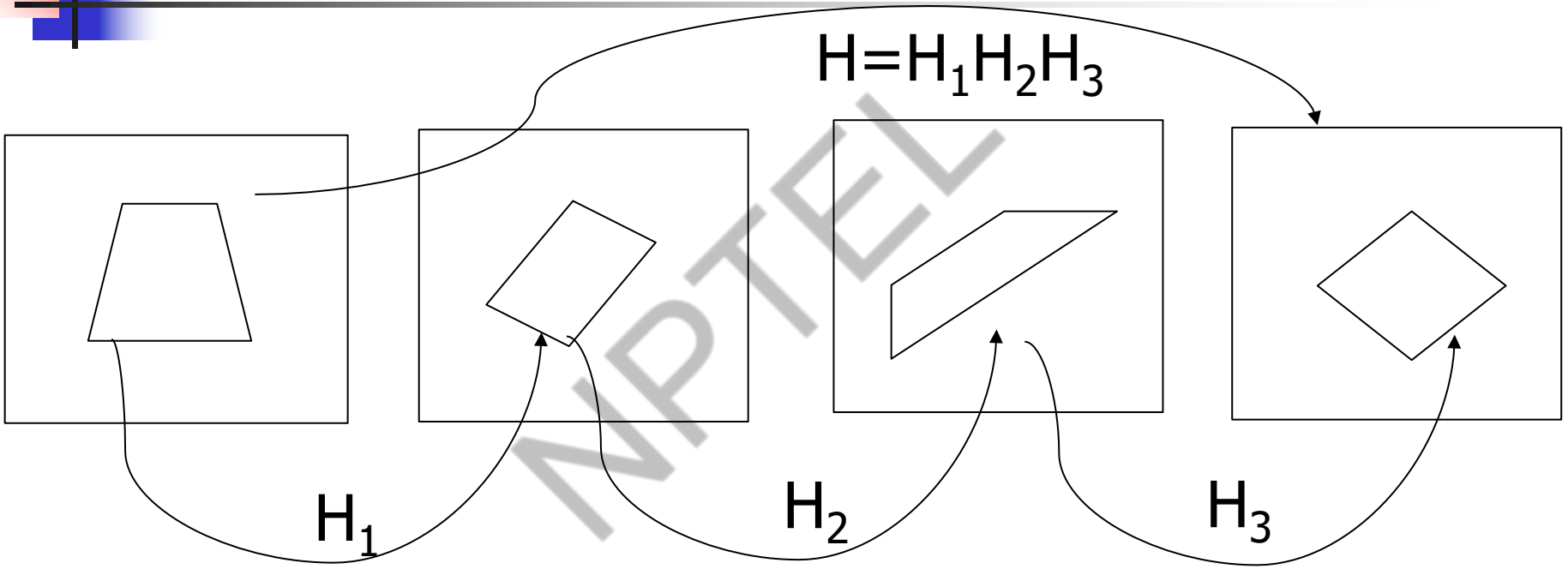
Projective linear group



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A cascade of transformation can be replaced by a single transformation.

Different compositions



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A series of transformation could be performed in different composition.

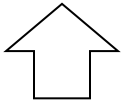
$$H=H_1(H_2H_3)=(H_1H_2)H_3$$

Subgroups and hierarchy

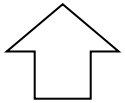
Projective linear group



Affine group (last row (0,0,1))



Euclidean group (upper left 2x2 orthogonal)

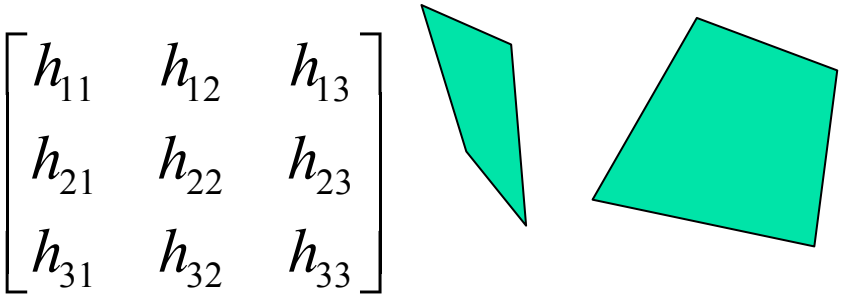


Oriented Euclidean group (upper left 2x2 det 1)

$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

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From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)



$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

Projective Group

$$\mathbf{x}' = \mathbf{H}_P \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix} \mathbf{x} \qquad \mathbf{v} = (v_1, v_2)^T$$

dof=8: 2 scale, 2 rotation, 2 translation, 2 line at infinity)

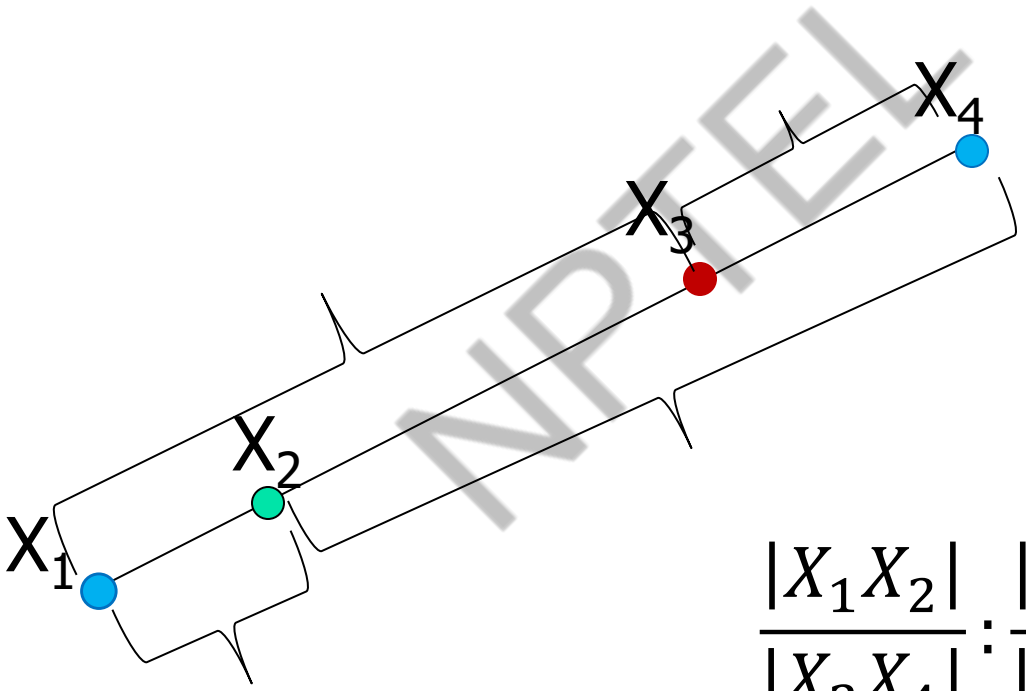
$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^T & v \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ v_1 x_1 + v_2 x_2 \end{pmatrix}$$

Line at infinity becomes finite, allows to observe vanishing points, horizon.

Concurrency, collinearity, order of contacts, cross ratio (ratio of ratio).

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Cross Ratio

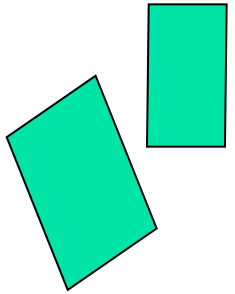


$$\frac{|X_1X_2|}{|X_2X_4|} : \frac{|X_1X_3|}{|X_3X_4|}$$

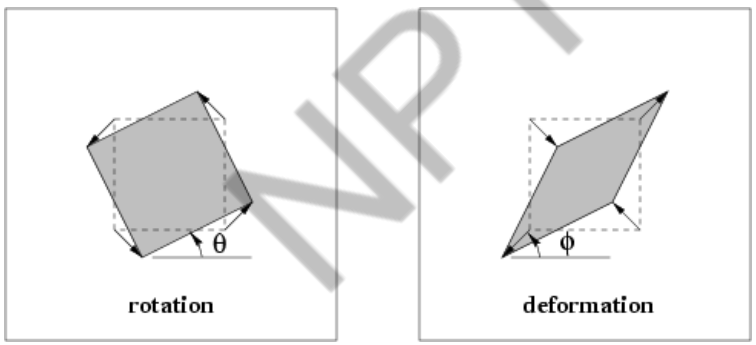
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Affine group

$$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^{\mathsf{T}} & 1 \end{bmatrix} \mathbf{x}$$



dof=6

$$\mathbf{A} = \mathbf{R}(\theta)\mathbf{R}(-\phi)\mathbf{D}\mathbf{R}(\phi)$$

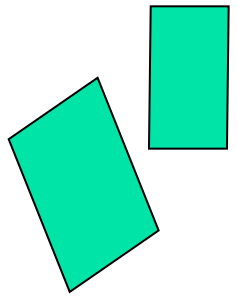
$$\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

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From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Affine group

$$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



$$\mathbf{x}' = \mathbf{H}_A \mathbf{x} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^{\mathsf{T}} & 1 \end{bmatrix} \mathbf{x} \qquad \text{dof=6}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^{\mathsf{T}} & v \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} \mathbf{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ 0 \end{pmatrix}$$

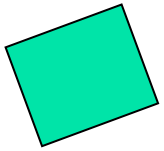
Line at infinity stays at infinity, but points move along line.

Parallelism, ratio of areas, ratio of lengths on parallel lines (e.g midpoints), linear combinations of vectors (centroids). **The line at infinity l_∞.**

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Similarity Group

$$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



$$\mathbf{x}' = \mathbf{H}_S \mathbf{x} = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \mathbf{x}$$

$$\mathbf{R}^\top \mathbf{R} = \mathbf{I}$$

dof=4 (1 scale, 1 rotation, 2 translation)

NPTEL

$$\mathbf{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \qquad \mathbf{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}$$

$$\mathbf{I}' = \mathbf{H}_S \mathbf{I} = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = se^{i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathbf{I}$$

Ratios of lengths, angles. **The circular points I,J.**

Isometry

$$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \varepsilon\cos\theta & -\sin\theta & t_x \\ \varepsilon\sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \qquad \varepsilon = \pm 1$$

Orientation preserving: $\varepsilon = 1$

Orientation reversing: $\varepsilon = -1$

dof=3 (1 rotation, 2 translation)

Invariants: length, angle, area

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Decomposition of projective transformations

$$\mathbf{H} = \mathbf{H}_S \mathbf{H}_A \mathbf{H}_P = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ 0^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & 0 \\ 0^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & 0 \\ \mathbf{v}^\top & v \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\top & v \end{bmatrix}$$

$$\mathbf{A} = s\mathbf{R}\mathbf{K} + \mathbf{t}\mathbf{v}^\top$$

$\mathbf{K}$ Upper-triangular

$\det \mathbf{K} = 1$ $\quad v \neq 0$

Decomposition unique (if chosen s>0)

QR decomposition:
Any square matrix decomposed as a product of an orthogonal matrix (Q) and an upper triangular matrix (R)

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Decomposition of projective transformations

$$\mathbf{H} = \mathbf{H}_S \mathbf{H}_A \mathbf{H}_P = \begin{bmatrix} s\mathbf{R} & \mathbf{t} \\ 0^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{K} & 0 \\ 0^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & 0 \\ \mathbf{v}^\mathsf{T} & v \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{v}^\mathsf{T} & v \end{bmatrix}$$

$$v \neq 0$$

$$\mathbf{A} = s\mathbf{R}\mathbf{K} + \mathbf{t}\mathbf{v}^\mathsf{T} \qquad \det \mathbf{K} = 1$$

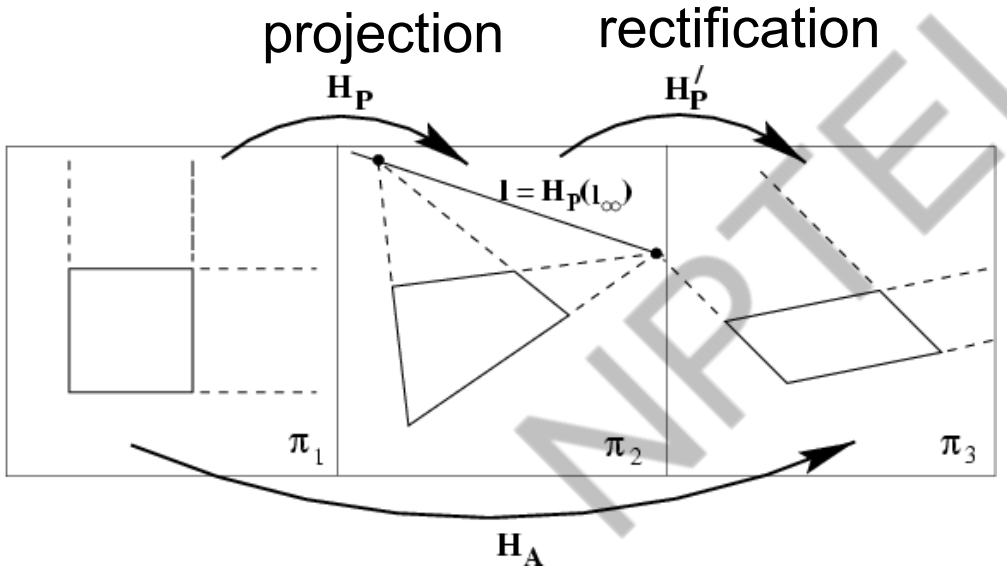
Example:

$$\mathbf{H} = \begin{bmatrix} 1.707 & 0.586 & 1.0 \\ 2.707 & 8.242 & 2.0 \\ 1.0 & 2.0 & 1.0 \end{bmatrix}$$

$$\mathbf{H} = \begin{bmatrix} 2\cos 45^\circ & -2\sin 45^\circ & 1.0 \\ 2\sin 45^\circ & 2\cos 45^\circ & 2.0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

NPTEL

Affine properties from images



$$l_\infty = [l_1 \quad l_2 \quad l_3]^T, l_3 \neq 0$$

$$H'_p = H_A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1 & l_2 & l_3 \end{bmatrix}$$

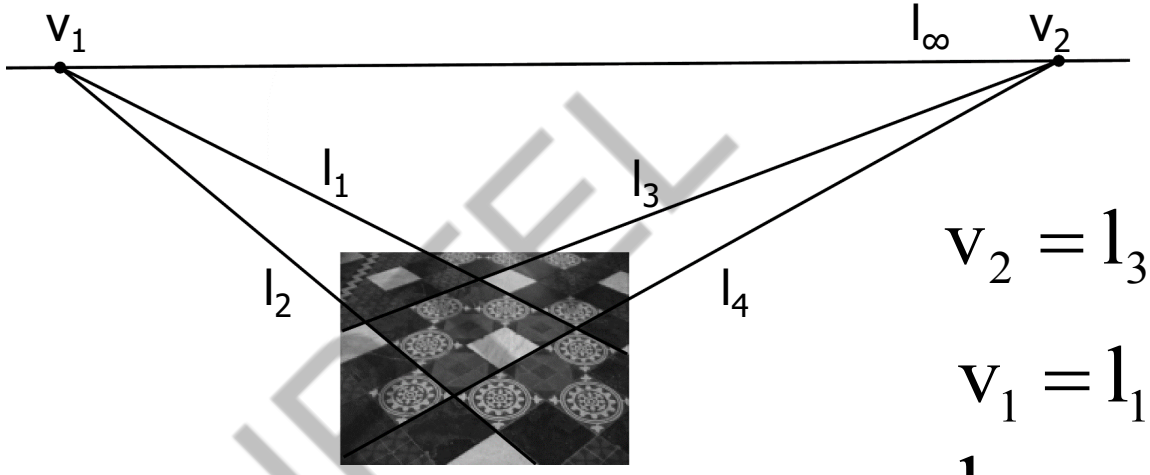
For any affine H_A.

$$H'^{-T}_p \begin{bmatrix} l_1 \\ l_2 \\ l_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

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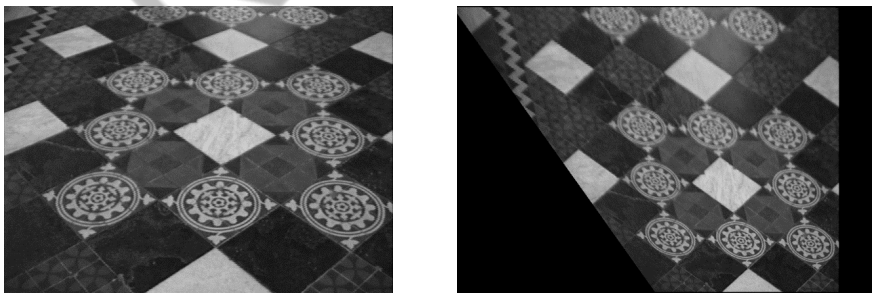
Affine rectification



$$v_2 = l_3 \times l_4$$

$$v_1 = l_1 \times l_2$$

$$l_\infty = v_1 \times v_2$$



From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

An example



(104,69,1) X (380, 71,1)

l_1=(-2, 276, -18836)

(122,226,1) x (366,228,1)

l_2=(-2, 244 , -54900)

(88,254,1) x (62,49,1)

l_3=(205,-26,-11436)

(390,250,1) x (406,53,1)

l_4=(197,16,-80830)

$v_1=l_1 \times l_2$ =(-10556416, -72128, 64)

$v_2=l_3 \times l_4$= (2284556,14317258,8402)

Vanishing Line: l_v= 1.0e+14 * (0,0.0009, -1.5097)

An example

Vanishing Line: l_v= 1.0e+14 * (0,0.0009, -1.5097)

Scaled l_v= (0, -0.0006,1)





Y=10^4/6

$$H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -.0006 & 1 \end{bmatrix}$$

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Conics in P^2

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

$$X^T C X = 0$$

$$C = \begin{bmatrix} a & \frac{b}{2} & \frac{d}{2} \\ \frac{b}{2} & c & \frac{e}{2} \\ \frac{d}{2} & \frac{e}{2} & f \end{bmatrix}$$

Conics identified by C with 5 d.o.f. $(a:b:c:d:e:f)$

A line tangent to the conic C satisfies $l^T C^* l = 0$

Dual conic

C^{-1}

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Transformation of conics under homography **H**

- $\mathbf{X'} = \mathbf{H}\mathbf{X}$
- $\mathbf{X}^T\mathbf{C}\mathbf{X} = 0$

  $\rightarrow (\mathbf{H}^{-1}\mathbf{X'})^T\mathbf{C}(\mathbf{H}^{-1}\mathbf{X'}) = 0$

  $\rightarrow \mathbf{X'}^T\mathbf{H}^{-T}\mathbf{C}\mathbf{H}^{-1}\mathbf{X'} = 0$

  $\rightarrow \mathbf{X'}^T\mathbf{C'}\mathbf{X'} = 0$

A conic remains a conic under homography.

where transformed conics $\mathbf{C'} = \mathbf{H}^{-T}\mathbf{C}\mathbf{H}^{-1}$

- $\mathbf{C'}^* = \mathbf{C'}^{-1} = (\mathbf{H}^{-T}\mathbf{C}\mathbf{H}^{-1})^{-1} = \mathbf{H}\mathbf{C}^{-1}\mathbf{H}^T$

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The circular points

$$\mathrm{I} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} \quad \mathrm{J} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} \quad \mathrm{I}' = \mathbf{H}_S \mathrm{I} = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = se^{i\theta} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathrm{I}$$

The circular points I, J are fixed points under the projective transformation **H** iff **H** is a similarity. They are also on $\mathbf{l}_\alpha$.

Every circle intersects $\mathbf{l}_\alpha$ at I and J.

Circle: $x_1^2 + x_2^2 + dx_1x_3 + ex_2x_3 + fx_3^2 = 0$

Setting x_3=0, $x_1^2 + x_2^2$=0. (I and J satisfies it)

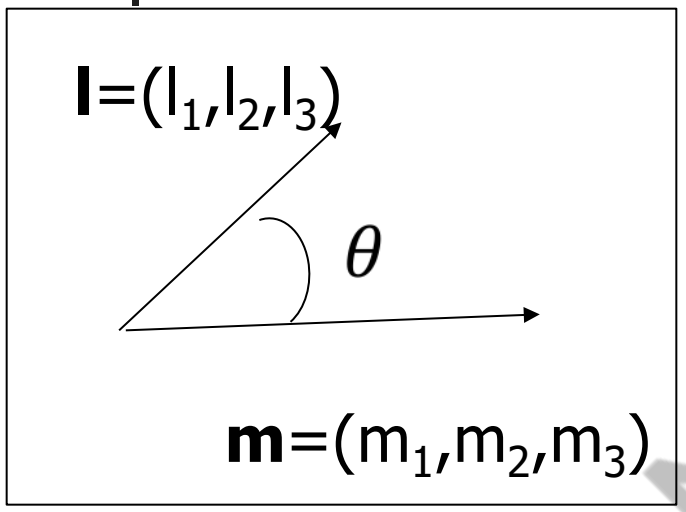
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Conic dual to the circular points (C_α^*)

- $C_\alpha^* = I.J^T + J.I^T$ (line conic)
- $C_\alpha^* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
- As I and J are fixed under similarity C_α^* is also fixed, i.e. $C_\alpha^{*\prime} = H_s\, C_\alpha^* H_s^T = C_\alpha^*$
- C_α^* is fixed iff H is a similarity.
- D.o.f. of transformed C_α^* is 4 and det. =0.
- l_α is the NULL vector of C_α^*.

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Measurement of angle under homography



$$cos(\theta) = \frac{l_1 m_1 + l_2 m_2}{\sqrt{(l_1^2 + l_2^2)(m_1^2 + m_2^2)}}$$

Invariant under homography

$$\cos(\theta) = \frac{\boldsymbol{l}^T C_\infty^* \boldsymbol{m}}{\sqrt{(\boldsymbol{l}^T C_\infty^* \boldsymbol{l})(\boldsymbol{m}^T C_\infty^* \boldsymbol{m})}}$$

Once $C_\alpha^{*\prime}$ is obtained Euclidean angle could be recovered.

If $\boldsymbol{l}$ and $\boldsymbol{m}$ orthogonal, $\boldsymbol{l}^T C_\alpha^{*\prime} \boldsymbol{m} = 0$.

$C_\infty^{*\prime} = H C_\infty^* H^T$ and $\boldsymbol{l}' = H^{-T}\boldsymbol{l}$

$$\boldsymbol{l}'^T C_\infty^{*\prime} \boldsymbol{m}'$$
$$= \boldsymbol{l}^T H^{-1} H C_\infty^* H^T H^{-T} \boldsymbol{m}$$
$$= \boldsymbol{l}^T C_\infty^* \boldsymbol{m}$$

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Estimation of $C_\alpha^{*\prime}$

- Use the property of orthogonal lines.
  - $\mathbf{l}^T\mathbf{C}_\alpha^{*\prime}\mathbf{m}=0$
- Minimum 5 such orthogonal pairs needed.
  - A typical equation

$$\begin{bmatrix} l_1m_1 & \frac{1}{2}(l_1m_2+l_2m_1) & l_2m_2 & \frac{1}{2}(l_1m_3+l_3m_1) & \frac{1}{2}(l_2m_3+l_3m_2) & l_3m_3 \end{bmatrix} C = 0$$

  - Where C represented by $(a,b,c,d,e,f)^T$.
- Apply direct linear transform (LSE method) to solve a set of homogeneous equations to get C.
- Make it ($C_\alpha^{*\prime}$) a rank 2 matrix using SVD on C.

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Recovery of metric properties

- Compute H from $C_\alpha^{*\prime}$ upto similarity.
  - Matrix decomposition method $C_\infty^{*\prime} = H C_\infty^* H^T$

$$C_\infty^{*\prime} = U \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T$$

H

- Apply H^{-1} to the image.

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Summary

- Various transformation under homography
  - $\mathbf{x'}=\mathbf{Hx}$
  - $\mathbf{l'}=\mathbf{H}^{-T}\mathbf{l}$
  - $\mathbf{C'}=\mathbf{H}^{-T}\mathbf{C}\mathbf{H}^{-1}$
  - $\mathbf{C'}^{*}=\mathbf{C'}^{-1}=\mathbf{H}\mathbf{C}^{-1}\mathbf{H}^{T}$
- Projective linear group, its subgroup and hierarchy
  - Projective linear group (8 D.O.F)
  - Affine group (6 D.O.F)
  - Euclidean group (4 D.O.F.)
  - Oriented Euclidean group (3 D.O.F)

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Summary (contd.)

- Conic dual to circular points (C_α^*)
  - Invariant under similarity transform.
  - l_α is the zero (NULL) vector.
  - Preserves cosine of angle of two lines under transformation

$$\cos(\theta) = \frac{\boldsymbol{l}^T C_\infty^* \boldsymbol{m}}{\sqrt{(\boldsymbol{l}^T C_\infty^* \boldsymbol{l})(\boldsymbol{m}^T C_\infty^* \boldsymbol{m})}}$$

- Use of homography
  - Affine rectification
  - Stratification (recovery of metric properties)

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